

Succulents

Plants with fleshy leaves or stems for storing water are known as succulents. They include the group of plants called cacti. There are three main types of succulent. Stem succulents, such as cacti, store water in their stems and tend to live in the driest climates. Leaf succulents, some of which are shown on this page, store water in their leaves and grow in damper conditions. Finally, root succulents have thickened roots that serve as water reservoirs.

LEAF SUCCULENTS

Leaf succulent plants live in semidesert and also in salt marshes, where the salty conditions mean that fresh water has to be carefully conserved. Their leaves wrinkle up in prolonged dry weather. When it rains, they swell again as the plant takes in water.

The whitish, waxy coating on the surface of the leaves of Senecio antandroi protects the plant from the harsh rays of the Sun

Senecio antandroi

The plump leaves of Haworthia cymbiformis are swollen with stored water

Haworthia cymbiformis

The necklace vine gets its name from its leaves, which look as if they have been threaded on a string

Necklace vine

Panda plant

Blue echeveria

Panda plant

Senecio antandroi

Cotyledon

Necklace vine

LEAVES THAT WITHSTAND DROUGHT

As much as nine-tenths of the weight of a succulent leaf may be stored water. To conserve this vital store, the leaves have a waxy surface that hinders transpiration, which is the process by which leaves lose water. Some succulent leaves have a woolly surface that helps to keep the leaf cool, which in turn reduces water loss. Succulent leaves have evolved in many unrelated plant species throughout the dry regions of the world.

FLEETING FLOWERS

Many desert plants are ephemerals. This means that they germinate only after rain, and then complete their life cycle very rapidly. For a few days after rain, the desert is ablaze with their flowers. This sea of little yellow flowers is made up of thousands of sand sunflowers, which have all come into bloom at once in the Utah Desert.

